

ATHUL BABURAJ

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SUMMARY

Solutions Architect with 3 years of enterprise cloud and AI platform experience at American Express, specializing in scalable solution design, governance frameworks, and cross-functional technical advisory. Track record of accelerating AI-powered platform adoption, driving measurable value realization, and architecting secure enterprise solutions—with hands-on depth in agentic AI systems and RAG-based knowledge platforms.

TECHNICAL SKILLS

Microsoft & AI Platforms: Microsoft 365, Copilot Studio, Azure, Power Platform (awareness), Microsoft Entra ID / Azure AD, SharePoint Online (conceptual), Teams extensibility concepts

AI & Agents: LLM-driven Agentic Workflows, RAG Architecture, Copilot Agent Patterns, Vector Search, Knowledge Graphs, Semantic Search, Recommendation Systems, LangChain, PyTorch

Cloud & DevOps: GCP, AWS, Kubernetes, OpenShift, Docker, Terraform, Helm, ArgoCD, Jenkins, CI/CD

Governance & Security: Cloud Compliance Dashboards, Billing Telemetry, Access Controls, Information Lifecycle, Security-by-Design Architecture

Programming: Python, Go, JavaScript, Java, Bash, SQL

Certifications: Google Cloud Professional Cloud Architect (PCA), Google Cloud Associate Cloud Engineer (ACE)

PROFESSIONAL EXPERIENCE

American Express, Bengaluru

Solutions Architect II

Mar 2026–Present

- Serve as trusted technical advisor to senior business and engineering stakeholders across product, platform, and security portfolios—translating business priorities into scalable AI solution architectures and driving enterprise-wide adoption
- Lead architecture design sessions for AI-powered productivity initiatives including an enterprise Recommendation Engine and RAG-based semantic Search platform, aligning designs with responsible AI principles and data governance requirements
- Own technical delivery for cross-portfolio onboarding programs (eBnC, Deep Instinct), developing stakeholder-specific roadmaps that accelerate time-to-value and resolve complex adoption blockers

Solutions Architect I

Aug 2023–Feb 2026

- Architected enterprise-scale platform migration for 100+ microservice portfolio, leading technical design reviews with application, security, and infrastructure teams and resolving deployment blockers through structured readiness assessments
- Built centralized governance and compliance dashboard aggregating billing APIs and platform telemetry, delivering executive-level visibility into cloud spend and reducing audit overhead by 60%—applying information protection and lifecycle governance principles at enterprise scale
- Developed reusable hybrid cloud connectivity patterns (AWS/On-premise to GCP) using Terraform, reducing platform onboarding time by 40% and enabling faster time-to-value for internal product teams
- Acted as technical liaison between internal platform consumers and engineering teams, enabling iterative improvement of shared services and driving sustained adoption

Software Engineer Intern

May 2022–Jun 2022

- Optimized Cassandra data models achieving 20% query latency reduction; built real-time migration tracking UI for Cloud Migration Orchestrator, improving stakeholder visibility into go-live readiness

AI & TECHNOLOGY PROJECTS

Vince OS: Local-First Agentic AI Assistant | github.com/athulbaburaj/project-vince

2025

- Designed and built a local-first agentic desktop system integrating LLM-driven workflows, vector memory, task orchestration, and knowledge graph visualization—demonstrating AI agent development, Copilot-pattern extensibility, and responsible deployment practices (Tauri, FastAPI, React)

Nomad-Zero: Distributed Process Orchestrator | github.com/athulbaburaj/nomad-zero

2025

- Built container orchestration platform in Go with gRPC control plane, gossip-based cluster membership, and work-stealing scheduling—demonstrating distributed systems design at infrastructure scale

Adaptive Control with Deep Reinforcement Learning (B.Tech Thesis) | doi.org/10.5281/zenodo.18047911

- Implemented DDPG-based AI agent with Actor-Critic architecture in PyTorch, achieving 68% improvement over classical controllers through simulation-driven optimization

EDUCATION

National Institute of Technology, Calicut

2019–2023

Bachelor of Technology in Electrical and Electronics Engineering | CGPA: 8.55/10.0 (First Class with Distinction)

ACHIEVEMENTS & LEADERSHIP

3rd Place, CaseCraft 2026 – IISc Bengaluru

Mar 2026

Developed go-to-market strategy for VoltSwap, demonstrating structured problem-solving and business case presentation

Joint Secretary, IEEE CS, NIT Calicut – Organized technical enablement workshops for 50+ participants 2021–22

Tech SIG Lead, Industrial Planning Forum – Coordinated technical conferences for 100+ participants 2020–21